

$\mathrm{Spec}(\mathbb{Z})$

arithmetic line

$(0) \bullet (p) = \langle p \rangle$ is the arithmetic analogue of a point.

Quotienting by (p) means: *move
to the fiber over the prime p .*

(5) 

(p) 



$$\mathbb{Z} \twoheadrightarrow \mathbb{Z}/(p) \cong \mathbb{F}_p$$

(3) 

(2) 